

9. The Tetrahedral Atlas

The closing remark of §8 is here promoted to a theorem. The root tetrahedron of the engine has three pairs of opposite edges: it is not one double-slit geometry but three, sharing a single frame, with the dynamic axis switching of FIG. 6 as the passage between them. The atlas of the Tetra Engine is a triple of Minkowski planes stitched along their seams — and the (S, T, U) round-trip proof of the coordinate system is Level 2 of this hierarchy running in code. This section makes each clause precise and verifies it against the engine's own chart data.

Fix the root tetrahedron T of the engine: the even-parity corners of the cube $[-1, 1]^3$,

$$V_1 = (1, 1, 1), V_2 = (1, -1, -1), V_3 = (-1, 1, -1), V_4 = (-1, -1, 1)$$

— the same parity rule that selects the FCC lattice sites, so that one principle builds both the coordinate system and the voxel (demo 22). T has six edges falling into three opposite pairs, one pair per coordinate axis:

X: V_1V_2 (in the plane $x = +1$) opposite V_3V_4 ($x = -1$) Y: V_1V_3 ($y = +1$) opposite V_2V_4 ($y = -1$) Z: V_1V_4 ($z = +1$) opposite V_2V_3 ($z = -1$)

Each opposite pair is a pair of skew lines. Nothing more is needed: each pair is a slit pair in the sense of §1.

D9 (The three charts). For $k \in \{X, Y, Z\}$, let ℓ_1^k, ℓ_2^k be the opposite-edge pair on axis k , $\mathcal{C}^k$ the congruence of their transversals, and M^k the resulting Minkowski plane (ray space $P^1 \times P^1$, with pencils and chains as in §§2–4). Parametrize the slits affinely by their endpoints — for chart X, $A(s) = (1-s)V_1 + sV_2$ and $B(t) = (1-t)V_3 + tV_4$ — and take U as the interpolation parameter along the ray:

$$P(s, t, u) = (1-u) A(s) + u B(t), (s, t, u) \in [0, 1]^3.$$

This is exactly the (S, T, U) map of FIG. 1 and demo 22, stated once per axis.

The engine of everything below is a one-line observation. Writing (w_1, w_2, w_3, w_4) for the barycentric coordinates of a point P with respect to $(V_1, \dots, V_4)$, chart X reads

$$P = (1-u)(1-s) V_1 + (1-u)s V_2 + u(1-t) V_3 + u t V_4,$$

so its (s, t, u) is the barycentric 4-tuple *regrouped by the opposite pair*: $u = w_3 + w_4$ is the total weight on slit B, and $s = w_2/(w_1+w_2)$, $t = w_4/(w_3+w_4)$ are the normalized weights within each slit. The other two charts are the other two regroupings. Geometrically: a tetrahedron is the join of any pair of its opposite edges — the union of all segments from one to the

other — and $\Delta^3 = \Delta^1 * \Delta^1$ in exactly three ways. The three charts are the three factorizations. (Verified: chart (s, t, u) matches the regrouped barycentric tuple to 10^{-14} in all three charts.)

Theorem 9 (Atlas Theorem).

1. **(Threading, three times.)** Each chart is a bijection between $(0, 1)^3$ and the interior of T ; every interior point carries three addresses, one per Minkowski plane. The center lock is barycentric: the centroid of T — the Cartesian origin — maps to $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ in *all three* charts, since $w = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ regroups the same way every time. (Verified: round trip exact to 10^{-14} per chart; center lock in all three.)
2. **(Edge-skeleton decomposition.)** In chart k , the six edges of T decompose as: the two seam edges of M^k , plus the four remaining edges, which are the four **corner rays** of the address square — the transversals with $(s, t) \in \{0, 1\}^2$. Consequently *the seam of each Minkowski plane consists of corner rays of the other two*: the planes are stitched to one another along their seams, each plane's singular locus sitting inside its neighbors' ray spaces as four ordinary (corner) addresses. The four vertices are the points where all three seams meet. (Proof: an edge $V_i V_j$ is the locus where the other two barycentric weights vanish; regrouped in a chart not pairing i with j , both normalized weights are 0 or 1. Verified for all 18 edge–chart incidences.)
3. **(Transition maps and the depth-transfer law.)** The transitions between charts are rational, with bilinear numerators and denominators — in barycentric terms they are trivial, being regroupings of the same 4-tuple. In particular the *depth* of each chart is bilinear in the *full address* of any other:

$$u_Y = (1-u_X) s_X + u_X t_X = \text{lerp}(s_X, t_X; u_X),$$

with the companion form $u_Z = (1-u_X) s_X + u_X (1-t_X)$ (the $1-t$ is an orientation convention of the slit parametrizations, not geometry). Depth in the next chart is the current chart's own interpolation rule applied to its address. The proof is one line: $u_Y = w_2 + w_4$. The remaining components come along for free: $s_Y (1-u_Y) = w_3$ and $t_Y u_Y = w_4$ are single barycentric weights. (Verified symbolically, and numerically for all six ordered chart pairs, residual $< 10^{-10}$.)

4. **(Dynamic axis switching is the atlas structure.)** Chart k 's inverse has denominators u and $1-u$ — equivalently, the axis- k coordinate at ± 1 — and within the closed tetrahedron these vanish *exactly on that chart's own seam edges*. On the interior of a seam edge, precisely one chart degenerates and the other two remain regular: the handoff of FIG. 6 always has somewhere to go. Only at the four vertices do all three charts degenerate at once — the points of the triple seam of part 2, the true corners of the atlas. The singularity-elimination claim of the patent draft (Claim 7) is, in this language, the statement that $\{M_X, M_Y, M_Z\}$ covers T minus its vertices with regular charts, and axis switching is the evaluation of the transition functions of part 3. (Verified: singular $\Leftrightarrow$ on

own seam, on 4000 random points of the closed simplex; exactly one singular chart on each seam-edge interior; all three singular at each vertex.)

Corollary 9.1 (Intrinsic stereo). Two addresses determine depth with no metric data. If a point has address (s_X, t_X) in chart X and (s_Y, t_Y) in chart Y , then applying the depth-transfer law in both directions gives two linear equations in (u_X, u_Y) , and

$$u_X = [s_Y + s_X (t_Y - s_Y)] / [1 - (t_X - s_X)(t_Y - s_Y)].$$

Depth in one Minkowski plane is a rational function of pure incidence data in two of them. (*Verified to 3×10^{-15} on random interior points.*) Note the contrast with Theorem 7: there, depth is recovered *within one* geometry, from the calibrated cross-ratio against F_1, F_2, S — a Level 3 statement. Here it is recovered *across two* geometries from addresses alone — a Level 2 statement, no rig constants anywhere in the formula. The dual cross-slit rig of demo 34 is the optical cousin of this corollary: it adjoins a second tetrahedron, rotated 90° about the shared axis, to gain an independent channel. The configuration differs — the dual rig's tetrahedra share an optical axis, while the atlas's charts share the frame itself, their seams meeting at the vertices — so the kinship is structural rather than an identity: in both cases a second Minkowski plane converts depth from a solved quantity into an incidence.

Remark (Level 2 running in code). Every formula in this section is built from barycentric weights and their ratios: affine invariants, Level 2 of §7, with the plane at infinity nominated and nothing else. No slit depth, no screen, no metric appears. The round-trip proof of the coordinate system (`tetra_validation.py`; §4 of the README) is therefore a certificate of exactly this level of the hierarchy: bijectivity off the seam is Axiom C1 chart by chart; exactness of the inversion is the affinity of U ; the center lock is the barycenter statement of Theorem 9.1. It certifies Level 2 and deliberately nothing above it — calibration (Level 3) enters only where the codec does, when the rig constants Z_v, Z_h turn the address into an aspect ratio and Theorem 7 takes over.

Honest caveat. Nothing in this section is deep: it is affine bookkeeping over Theorem 1, and its entire content is the combinatorics of the 3-simplex — three opposite-edge pairs, three join factorizations, three regroupings of one barycentric 4-tuple. Its value is that the bookkeeping is *the engine's own*: the chart data verified here is copied from demo 22 and FIG. 6, so the identification of the Tetra Engine's atlas with a triple of stitched Minkowski planes is a statement about shipped code, not about an idealization. The corner rays of part 2 (corners of the bounded address square) should not be confused with the corner ray of the projective screen chart in Theorem 3.2; they are different points of the ray space.

Verification: `atlas_checks.py` confirms all parts — barycentric regrouping (10^{-14}), per-chart round trips and triple center lock (10^{-14}), the 18 edge-chart incidences of the skeleton

decomposition, the depth-transfer law symbolically and for all six ordered chart pairs (10^{-10}), the seam-exact singularity pattern of the axis switch on 4000 random simplex points, and the intrinsic-stereo formula (3×10^{-15}). 15 checks, 15 passing.